



Hashing, Homework 5

1 Universal Hash Family.

1. Let $U = \{1, 2, 3, 4\}$, and consider a table at size $m = 2$. Find 4 functions $h_1, h_2, h_3, h_4 : U \rightarrow [m]$ such that the family $\mathcal{H} = \{h_1, h_2, h_3, h_4\}$ is universal.
2. Give an example of a family of functions $\mathcal{H} = \{h : U \rightarrow [m]\}$ such that for any $x \in U$ and $j \in [m]$, on the one hand:

$$\Pr[h(x) = j] \leq \frac{1}{m}$$

Yet, on the other hand, $\mathcal{H}$ is not universal.

3. U is the set of all matrices $\mathbb{F}_2^{n \times n}$ and $h_X : U \rightarrow \{0, 1\}$, $h_X(A) = \text{Tr}(XA)$. Prove that $\mathcal{H} = \{h_X : X \in \mathbb{F}_2^{n \times n}\}$ is a universal family.

2 Universal Hashing

1. Prove by yourself the following claim.¹

Claim 1. Let $\mathcal{H} = \{h : U \rightarrow [m]\}$ be a universal hash family. Denote by T an m -cell table, and suppose we hash n items $x_1, \dots, x_n$ into T . Then, for any x_i , the expected number of items that are hashed to the same cell as x_i is:

$$1 + \frac{n-1}{m}$$

2. Let U be a set of keys. Prove the following claim.

Claim 2. When $|U| \geq m \cdot n$, for every hash table of size m that uses the chaining method, there exists a subset $S \subset U$ of size n such that every element in S is mapped to the same table cell.

¹Given and proven in both the lecture and the recitation.